# Op-Amp ECG Heartbeat Detector

Precision rectifier · peak detector · Schmitt comparator

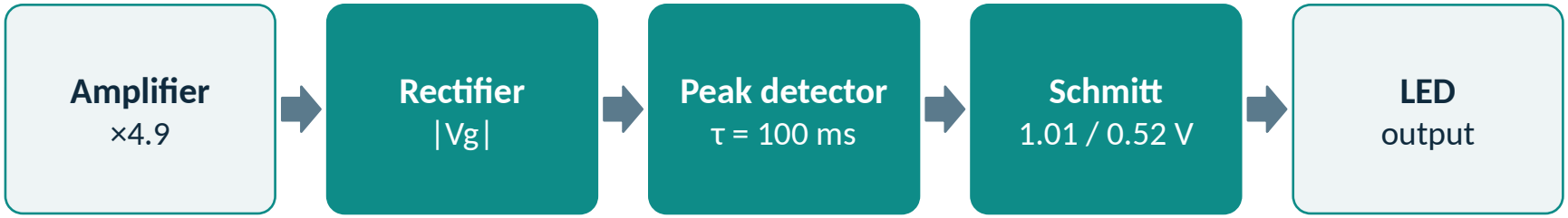


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Nonlinear Circuits · Universidad de Ibagué · 2026

# Problem

Design and implement an op-amp-based heartbeat detector powered from ±12 V, able to identify every QRS complex of an ideal 0.5 to 2 Vpp ECG-like signal regardless of its polarity or waveform type, delivering one clean rectangular pulse per beat over a range of 40 to 150 beats per minute.

# One pulse per beat, regardless of polarity

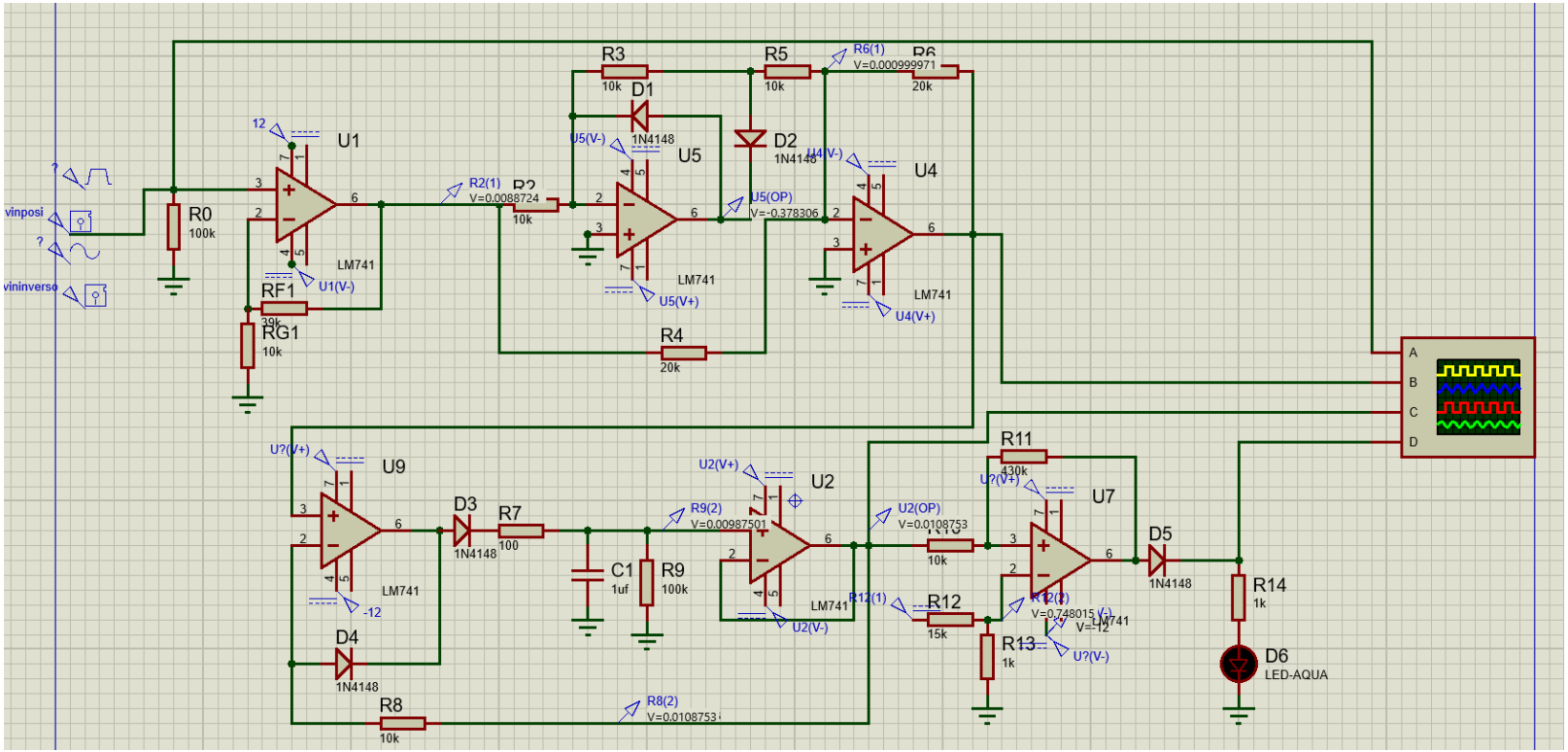



*Nonlinear blocks*

## Why a linear circuit is not enough

- The R wave can be positive or negative → rectify
- The QRS has several lobes → merge them into one
- Design limits: 0.5 to 2 Vpp

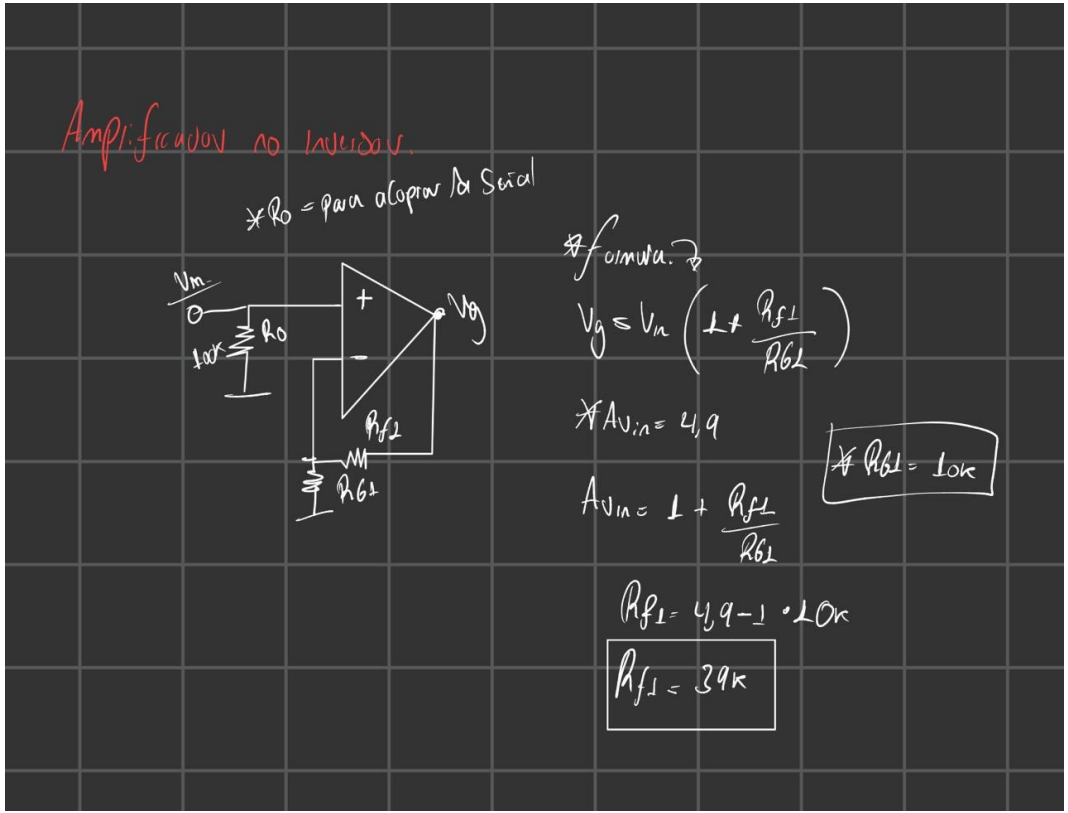
# Full circuit: 6 op-amps at ±12 V




Simulated in Proteus with LM741

# Stage 1 · Non-inverting amplifier

1




## Key points

- Vg = Vin (1 + Rf1/Rg1)
- Av = 4.9
- Rg1 = 10 kΩ → Rf1 = 39 kΩ
- 0.5–2 Vpp → 2.45–9.8 V

**Av = 4.9**

valid range: 4.04 to 5.25

La ganancia se determina por la saturación del Schmitt al tener una alimentación de ± 12v

$V_{sat} \approx 10,5v$, Pq el op usa en Vx y Podemos dejar un margen

$2 Im_{max}$

$2 v_{pp}$

$6 \cdot 2V \leq 10,5v$

$6 \leq 5,25v$

} Para q no se recorte la señal

Entonces la ganancia Sera de 4,9 para dejar un margen.

Comprobación max

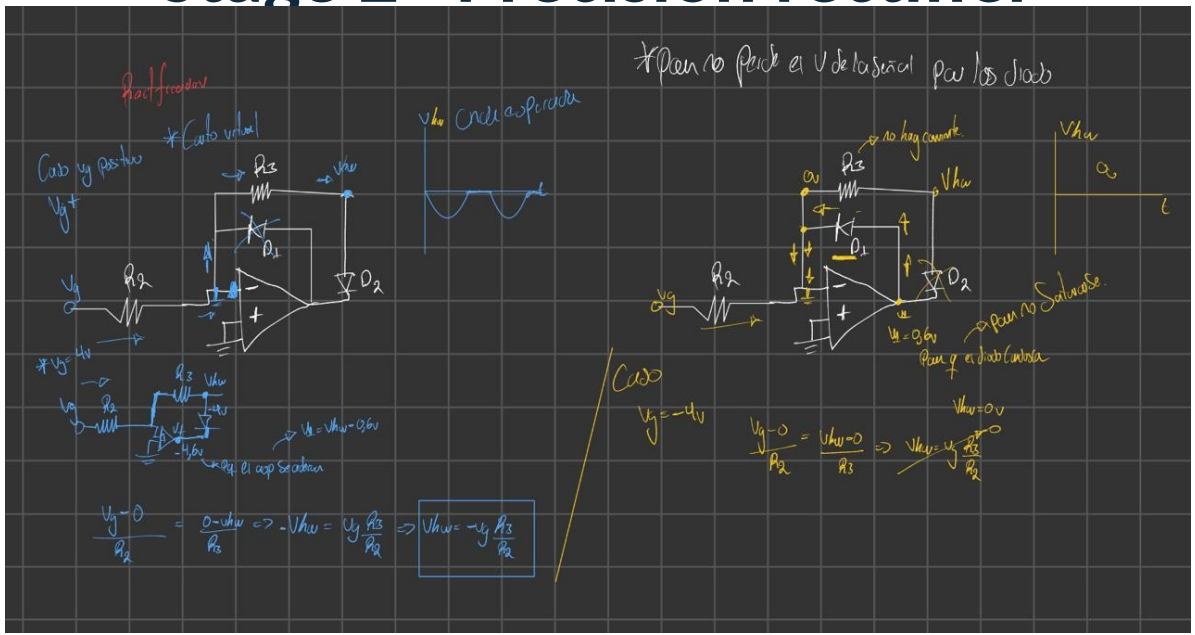
$2 \cdot 4,9 = 9,8v$ y $\boxed{9,8v < 10,5}$ Cumple

Comprobación min

$0,5 \cdot 4,9 = 2,45v$ y $2,45v < 10,5v$

# Stage 2 · Precision rectifier

2

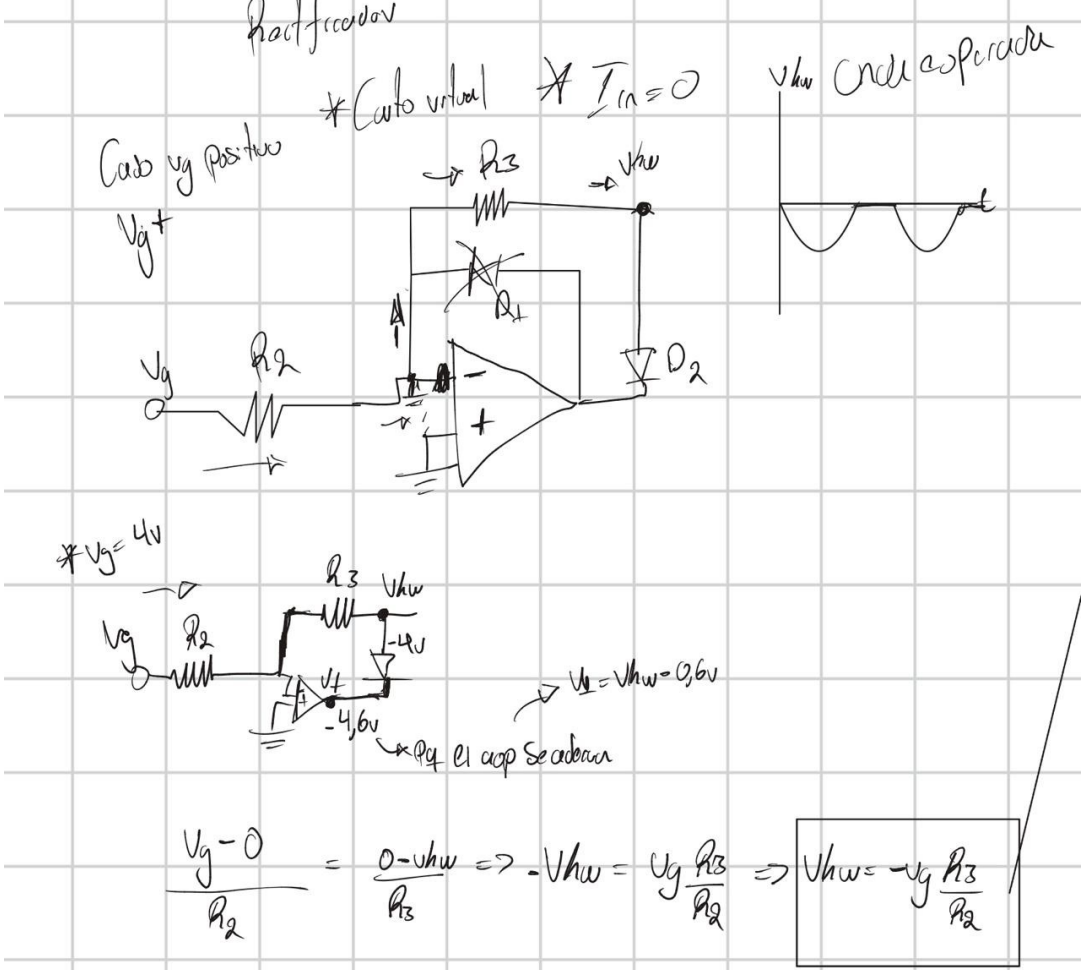



**Vg > 0**
D2 conducts · Vhw = −Vg

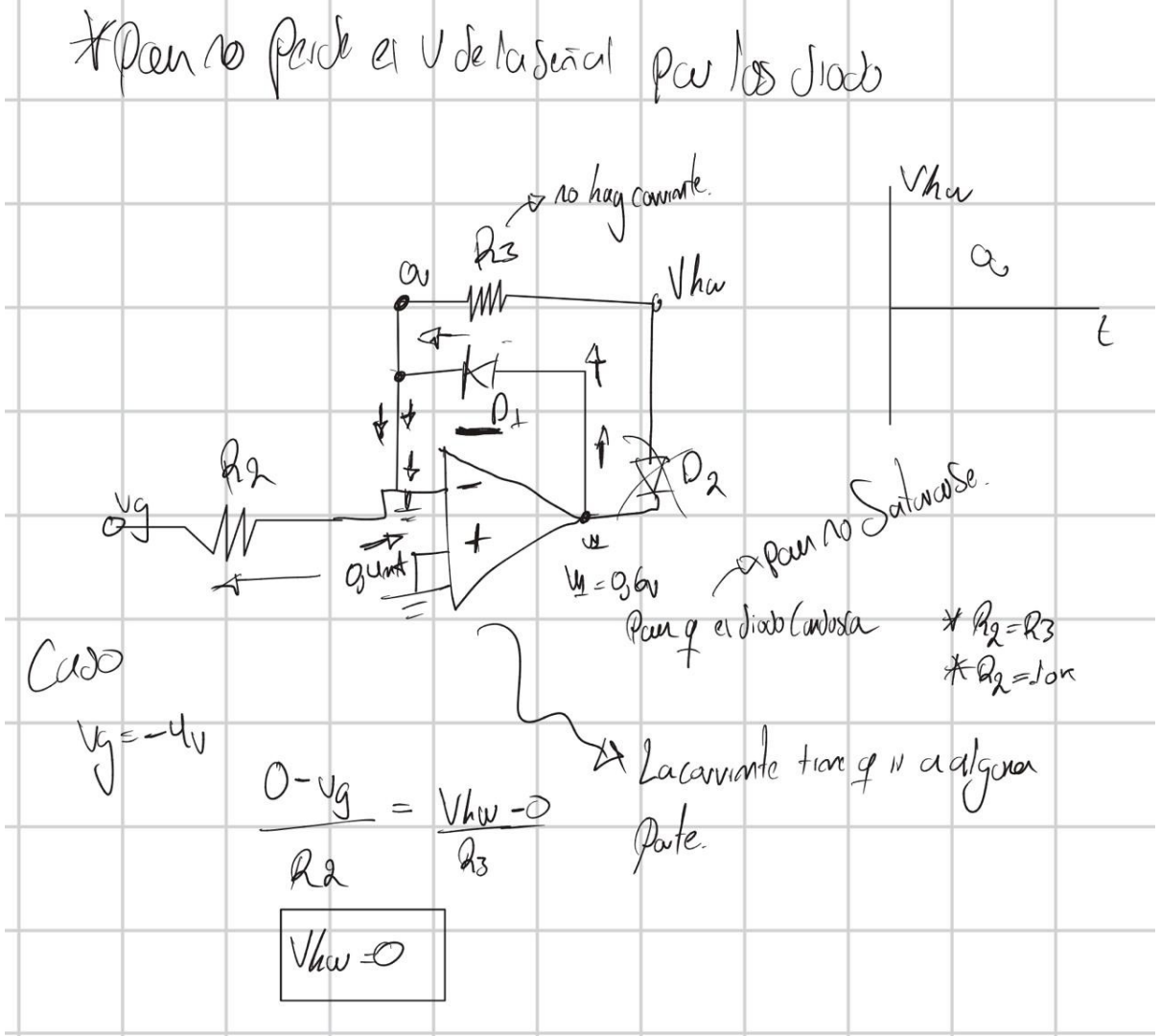
**Vg < 0**
D1 conducts · Vhw = 0

**Precision**
Dead zone ≈ 3 µV

Case 1, Vg +

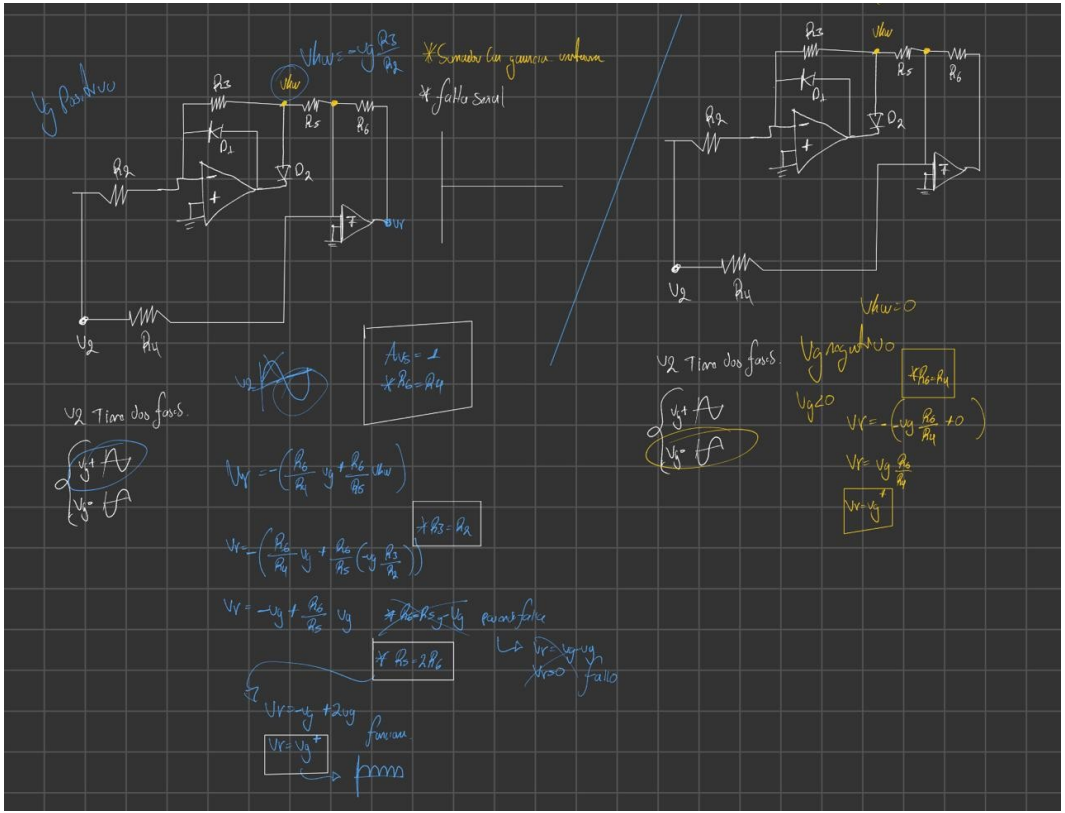

Ractfeedback
* Cuto virtual
* $I_{in} = 0$
Vhw onda esperada
Caso vg positivo
Vg+
R3
Vhw
D1
D2
Vg
R2
* Vg = 4V
-4V
Vi
-4,6V
$V_i = V_{hw} = 0,6V$
* Pq el aop se adecua
$\frac{V_g - 0}{R_2} = \frac{0 - V_{hw}}{R_3} \Rightarrow -V_{hw} = V_g \frac{R_3}{R_2} \Rightarrow V_{hw} = -V_g \frac{R_3}{R_2}$

Case 2, Vg -


*Para no perder el V de la señal para los diodo
no haya corriente.
R3
Vhw
D1
D2
R2
Vg
Vhw
t
Para no Saturarse.
Vy = 0,6v
Para q el diodo conduzca
* R2 = R3
* R2 = 10k
Caso
Vg = -4V
0 - Vg / R2 = Vhw - 0 / R3
La corriente tiene q ir a alguna parte.
Vhw = 0

# Stage 2 · Summer: from half-wave to |Vg|

2

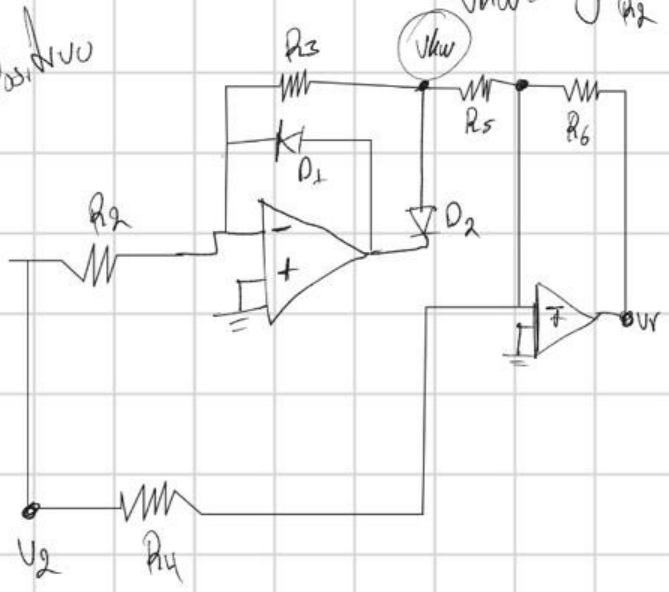


## Key points

- Vr = −(Vg + 2·Vhw)
- R6 = R4 = 20 kΩ
- R6 = 2·R5 (R5 = 10 kΩ)
- Result: Vr = |Vg|

**1 %**

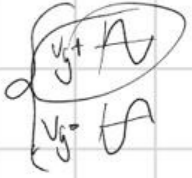
tolerance for R2 to R6




$V_{hw} = -V_g \frac{R_3}{R_2}$

* Sumador con ganancia unitaria
* falla Serial

U2 Tiene dos fases.







$A_{VS} = 1$
$* R_6 = R_4$

$$V_r = -\left(\frac{R_6}{R_4} V_g + \frac{R_6}{R_5} V_{hw}\right)$$

$* R_3 = R_2$

$$V_r = -\left(\frac{R_6}{R_4} V_g + \frac{R_6}{R_5}\left(-V_g \frac{R_3}{R_2}\right)\right)$$

$$V_r = -V_g + \frac{R_6}{R_5} V_g$$

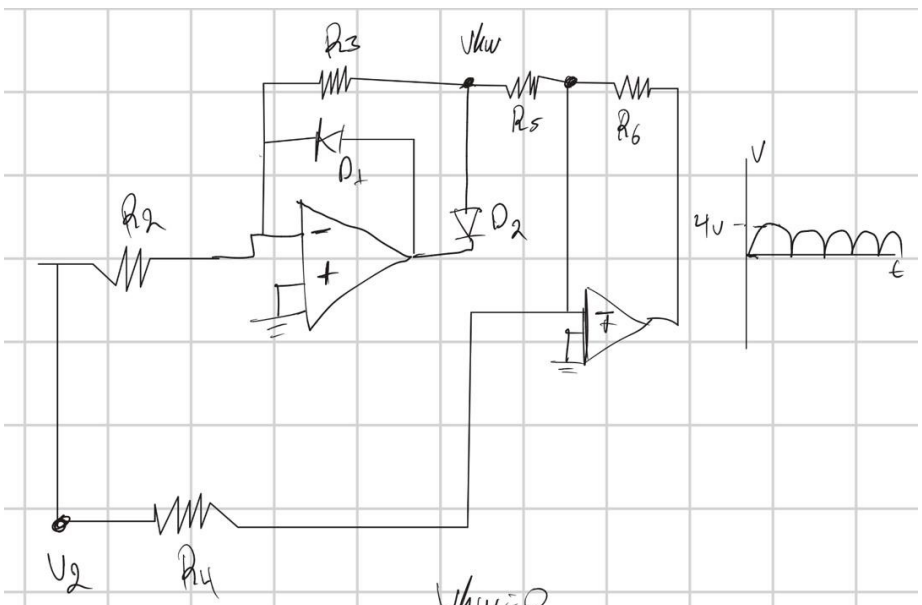
~~$* R_6 = R_5 g - V_g$~~ para una falla

$V_r = V_g - V_g$
~~$V_r = 0$~~ falla

$* R_5 = 2R_6$

$V_r = -V_g + 2V_g$

$V_r = V_g +$ funciona.




Vhw=0

U2 Tiene dos fases.

Vg negativo

$R_6 = R_4$

Vg+

Vg-

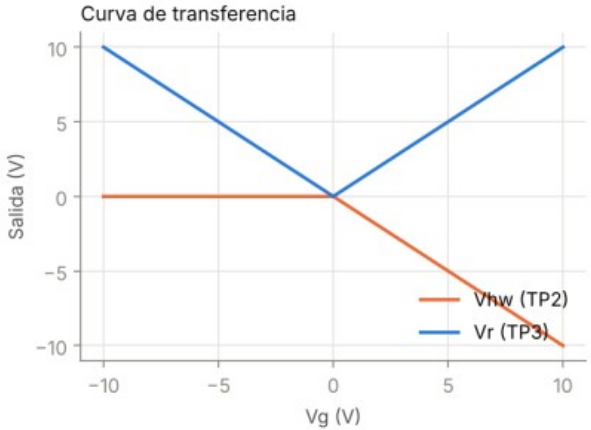
Vg<0

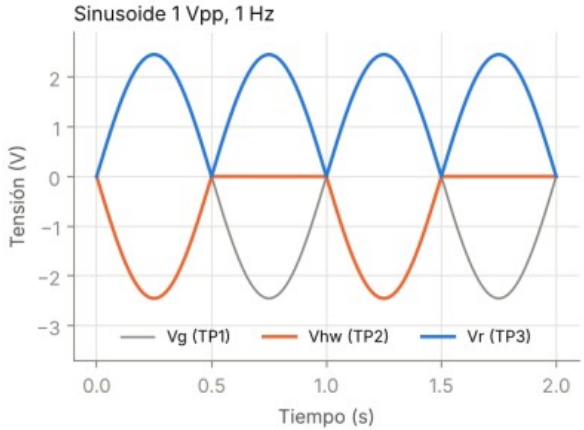
$$Vr = -\left(-Vg\frac{R_6}{R_4} + 0\right)$$

$$Vr = Vg\frac{R_6}{R_4}$$

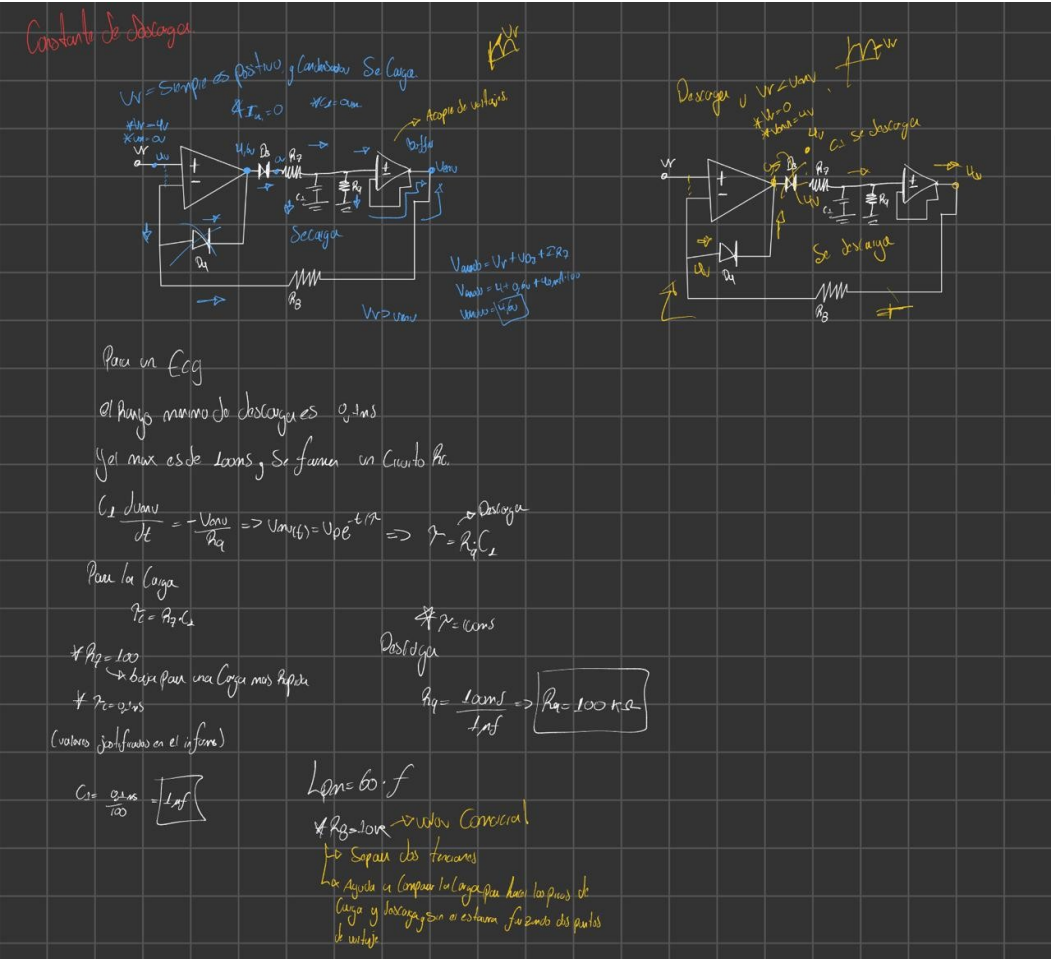
$$Vr = Vg^{+}$$

| | $V_g = +4\,\mathrm{V}$ | $V_g = -4\,\mathrm{V}$ |
|---|---|---|
| Salida de U2 | $-4{,}6\,\mathrm{V}$ | $\approx +0{,}62\,\mathrm{V}$ |
| D1 / D2 | Corte / conduce | Conduce / corte |
| $V_{hw}$ (TP2) | $-4\,\mathrm{V}$ | $0\,\mathrm{V}$ |
| Corriente por D2 | 0,8 mA (R3 + R5) | 0 |
| $V_r$ (TP3) | $+4\,\mathrm{V}$ | $+4\,\mathrm{V}$ |



# Stage 3 · Peak detector




## Key points

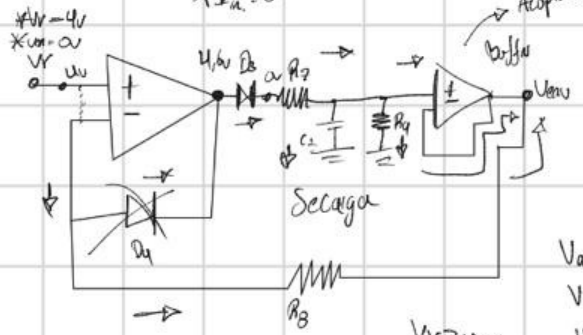
- τ = R9·C1 = 100 kΩ · 1 µF = 100 ms
- Valid range: 90–110 ms
- Buffer: τ is independent of R8

**100 ms**
discharge time constant

**0.1 ms**
charge time constant

# Constante de Descarga

$V_r$ = Siempre es positivo, y Condensador Se Carga




$I_{in} = 0$ $\quad C_1 = 0\mu$

$V_{salida} = V_r + V_{D3} + IR_7$

$V_{salida} = 4 + 0{,}6v + 40mA \cdot 100$

$V_{salida} = 4{,}6v$

$V_r > V_{sal}$

Para un Ecg

El Rango maximo de descarga es 0,1ms

y el max es de 100ms, Se forman un Circuito RC.

$$C_1 \frac{dV_{sal}}{dt} = -\frac{V_{sal}}{R_q} \Rightarrow V_{sal}(t) = V_p e^{-t/\tau} \Rightarrow \tau = R_q \cdot C_1$$

($\tau$: Descarga)

Para la Carga

$\tau_c = R_7 \cdot C_1$

$R_7 = 100$ → baja para una Carga mas Rapida

$\tau_c = 0{,}1ms$

(Los justificados en el informe)

$$C_1 = \frac{0{,}1ms}{100} = 1\mu f$$

$\tau = 100ms$

Descarga

$$R_q = \frac{100ms}{1\mu f} \Rightarrow R_q = 100\,k\Omega$$

$Lpm = 60 \cdot f$

$R_8 = 10k$ → valor Comercial

↳ Sopesa las tensiones

↳ Ayuda a Comparar la Carga para hacer los picos de Carga y Descarga sin el estaria forzando dos puntos de voltaje.

**Límite inferior de $\tau$.** El QRS no es un pulso único: sus lóbulos (Q, R y S) están separados unos 20 ms y el complejo completo dura de 80 a 120 ms en un adulto [4]. Entre lóbulos la envolvente no debe perder más del 20 % (criterio de diseño), para que el comparador no se libere en medio del complejo y cuente un latido dos veces:

$$e^{-20\,\mathrm{ms}/\tau} > 0{,}8 \;\Rightarrow\; \tau > \frac{20\,\mathrm{ms}}{\ln(1/0{,}8)} \approx 90\,\mathrm{ms} \tag{16}$$

**Condición de descarga.** La condición para el límite superior se deduce siguiendo un latido completo:

**Límite superior de $\tau$.** El peor caso es el pico más alto (9,8 V, que tarda más en descargarse) a la frecuencia más alta del enunciado (150 lpm, $T = 400\,\mathrm{ms}$). Con $V_{TL} = 0{,}52\,\mathrm{V}$ (etapa 4) en la ecuación (17):

$$75\,\mathrm{ms} + \tau \ln \frac{9{,}8}{0{,}52} < 400\,\mathrm{ms} \;\Rightarrow\; \tau < \frac{325\,\mathrm{ms}}{2{,}94} \approx 110\,\mathrm{ms} \tag{18}$$

Se elige el punto medio de la ventana 90 ms to 110 ms, $\tau = 100\,\mathrm{ms}$.

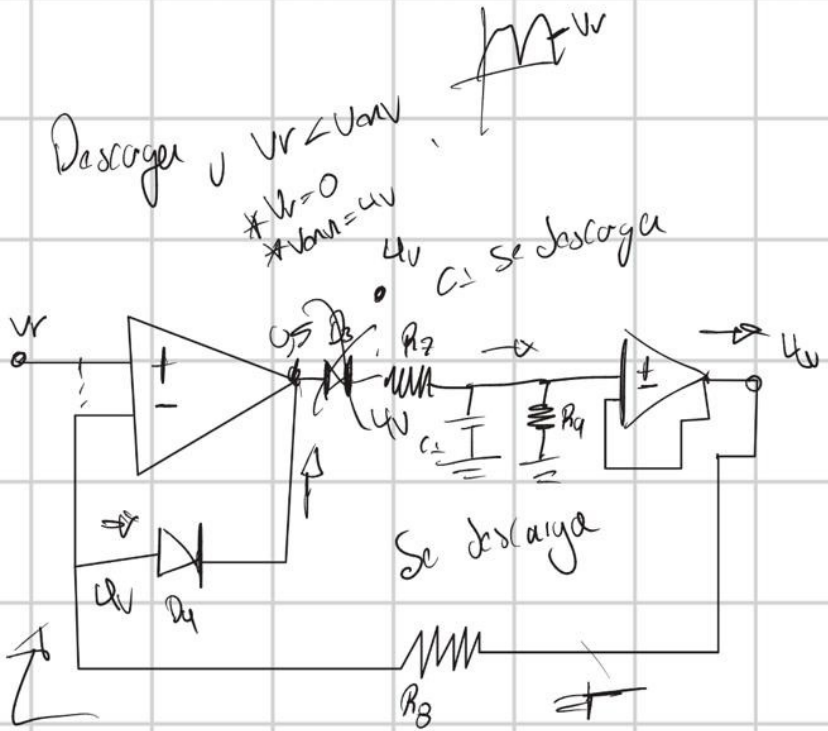
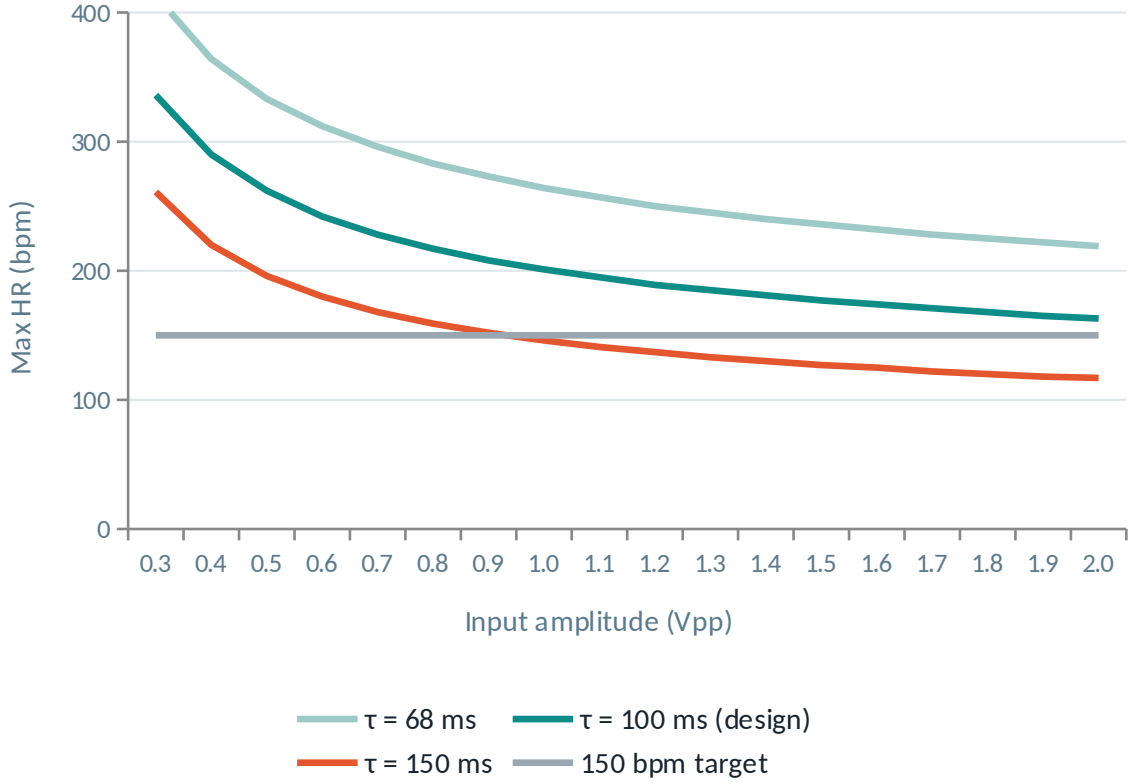

Vr
Descarga V Vr < Vbav
*Vr = 0
*Vbav = 4V
4V
C1 se descarga
Vr
D3
R7
4V
C1
R9
Vo
Se descarga
4V
D4
R8

**Tabla 5:** Datos de partida del detector de pico.

| Dato | Valor | Origen |
|---|---|---|
| Pico de $V_r$ | 2,45 a 9,8 V | Entrada de 0,5 a 2 $V_{pp}$ por la ganancia de 4,9 |
| Ancho del pulso ($w$) | 75 ms | Pulso de prueba, similar a un QRS |
| Separación entre lóbulos del QRS | $\approx 20$ ms | Tiempo típico entre las ondas R y S |
| Caída máxima entre lóbulos | 20 % | Criterio de diseño |
| Frecuencia cardiaca máxima | 150 lpm ($T = 400$ ms) | Rango de diseño |
| Umbral de liberación | $V_{TL} = 0{,}52$ V | Etapa 4 |
| Corriente máxima del op-amp | $\approx 20$ mA | Hoja de datos |
| Corriente de polarización | $\approx 80$ nA | Hoja de datos del 741 (peor caso) |

# τ sets the maximum heart rate




## Formula (taken from the report)

$$FC_{max} = \frac{60}{w + \tau \ln(V_p/V_{TL})} \text{ lpm}$$

**163 bpm**

limit at 2 Vpp

$$t_{desc} = \tau \ln \frac{V_p}{V_{TL}} \tag{11}$$

**Tabla 6:** Tiempo de descarga y ancho del pulso de salida.

| Entrada | $V_p$ | $t_{desc}$ | Ancho de salida $(w + t_{desc})$ |
|---|---|---|---|
| 0,5 $V_{pp}$ | 2,45 V | 154 ms | 229 ms |
| 1 $V_{pp}$ | 4,90 V | 224 ms | 299 ms |
| 2 $V_{pp}$ | 9,80 V | 293 ms | 368 ms |

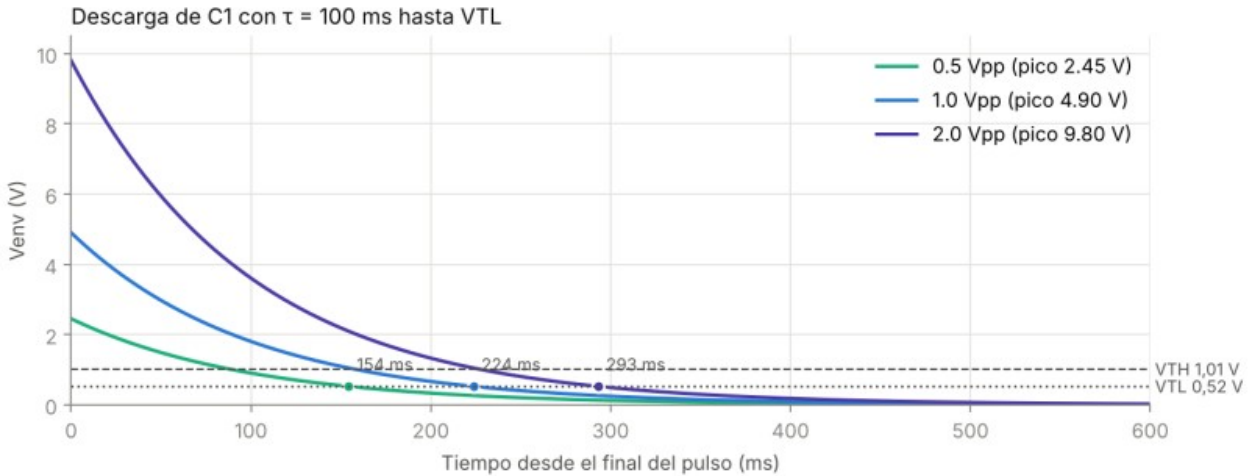
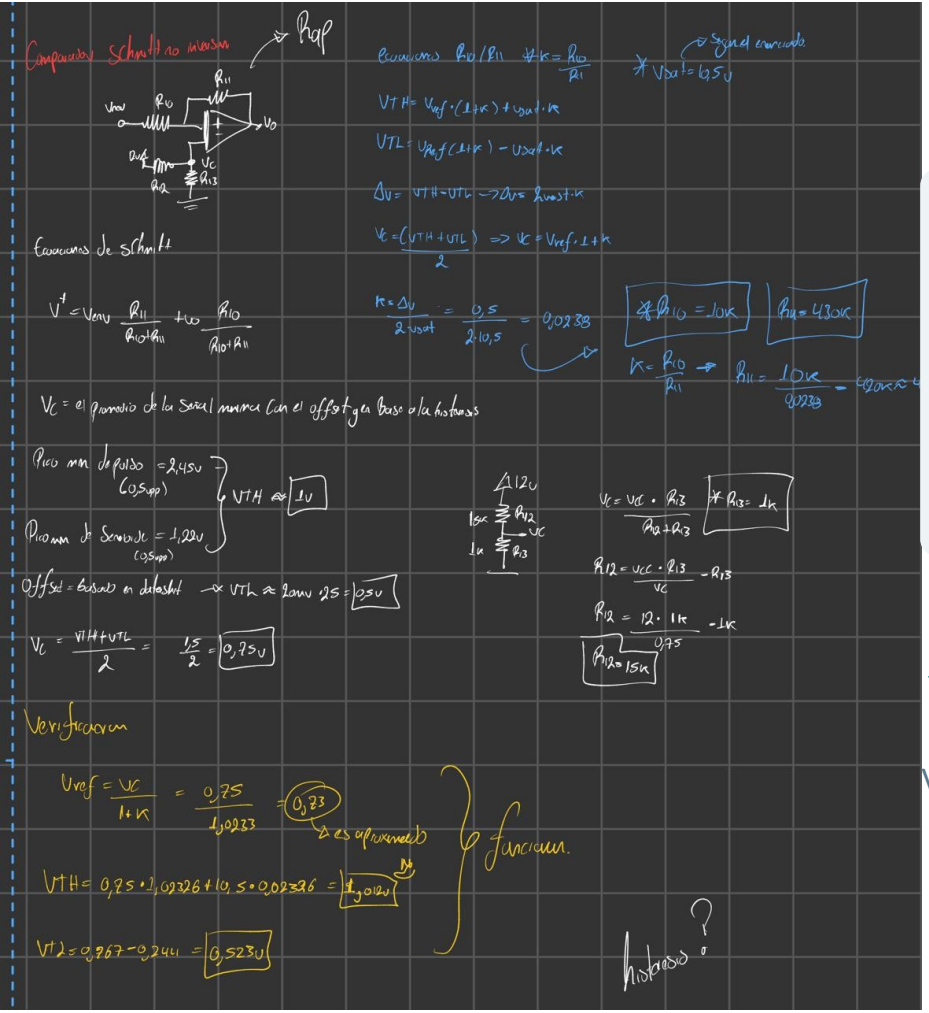



**Figura 7:** Descarga de C1 hasta $V_{TL}$ para las tres amplitudes.

# Stage 4 · Schmitt comparator




## Key points

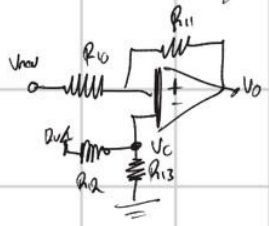
- k = R10/R11 = 0.0233 (430 kΩ)
- Vref = 0.75 V (R12 = 15 kΩ, R13 = 1 kΩ)

**1.01 V**
VTH (trigger)

**0.52 V**
VTL (release)

**0.49 V**
hysteresis

Comparador Schmitt no inversor → Rap




Ecuaciones R10 / R11 ✱ $k = \frac{R10}{R11}$ ✱ $V_{sat} = 10{,}5V$ → Señal cuadrada

$V_{TH} = V_{ref} \cdot (1+k) + V_{sat} \cdot k$

$V_{TL} = V_{ref}(1+k) - V_{sat} \cdot k$

$\Delta V = V_{TH} - V_{TL} \rightarrow \Delta V = 2 V_{sat} \cdot k$

$V_c = \frac{(V_{TH} + V_{TL})}{2} \Rightarrow V_c = V_{ref} \cdot 1 + k$

Ecuaciones de Schmitt

$$V^t = \left(V_{ent} \frac{R11}{R10+R11} + V_o \frac{R10}{R10+R11}\right)\left(\frac{R10+R11}{R11}\right)$$

$k = \frac{\Delta V}{2 \cdot V_{sat}} = \frac{0{,}5}{2 \cdot 10{,}5} = 0{,}0238$

✱ $R_{10} = 10k$ | $R_{11} = 430k$

$k = \frac{R10}{R11} \rightarrow R11 = \frac{10k}{0{,}0238} = 420k \approx 430k$

$V_c$ = el promedio de la señal cuadrada con el offset en base a la hoja de datos

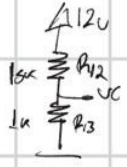
Pico mín de pulso = 2,45V (0,5 Vpp)

Pico máx de Senoide = 1,22V (0,5 Vpp)

} $V_{TH} \approx$ 1V

Offset = Basado en datasheet → $V_{TH} \approx 20mV \cdot 25 =$ 0,5V

$V_c = \frac{V_{TH}+V_{TL}}{2} = \frac{1{,}5}{2} =$ 0,75V → Tiene que estar por encima del vuelo.




$V_c = \frac{V_{CC} \cdot R13}{R12+R13}$ ✱ $R_{13} = 1k$

$R12 = \frac{V_{CC} \cdot R13}{V_C} - R13$

$R12 = \frac{12 \cdot 1k}{0{,}75} - 1k$

$R_{12} \approx 15k$

Verificacion

$$V_{ref} = \frac{V_C}{1+K} = \frac{0,75}{1,0233} = 0,73$$

↳ es aproximado

$$V_{TH} = 0,75 \cdot 1,02326 + 10,5 \cdot 0,02326 = 1,012V$$

$$V_{TL} = 0,767 - 0,244 = 0,523V$$

} funciona.

Umbrales de histeresis

$-10,5V$ (espera latido) $= V_{TH}$

$+10,5V$ (Latido detectado) $= V_{TL}$

$$V_{TH} = 0{,}750(1{,}02326) + 10{,}5(0{,}02326) = 0{,}767 + 0{,}244 = 1{,}012\,\mathrm{V} \tag{20}$$

$$V_{TL} = 0{,}767 - 0{,}244 = 0{,}523\,\mathrm{V}, \qquad \Delta V = 0{,}488\,\mathrm{V} \tag{21}$$

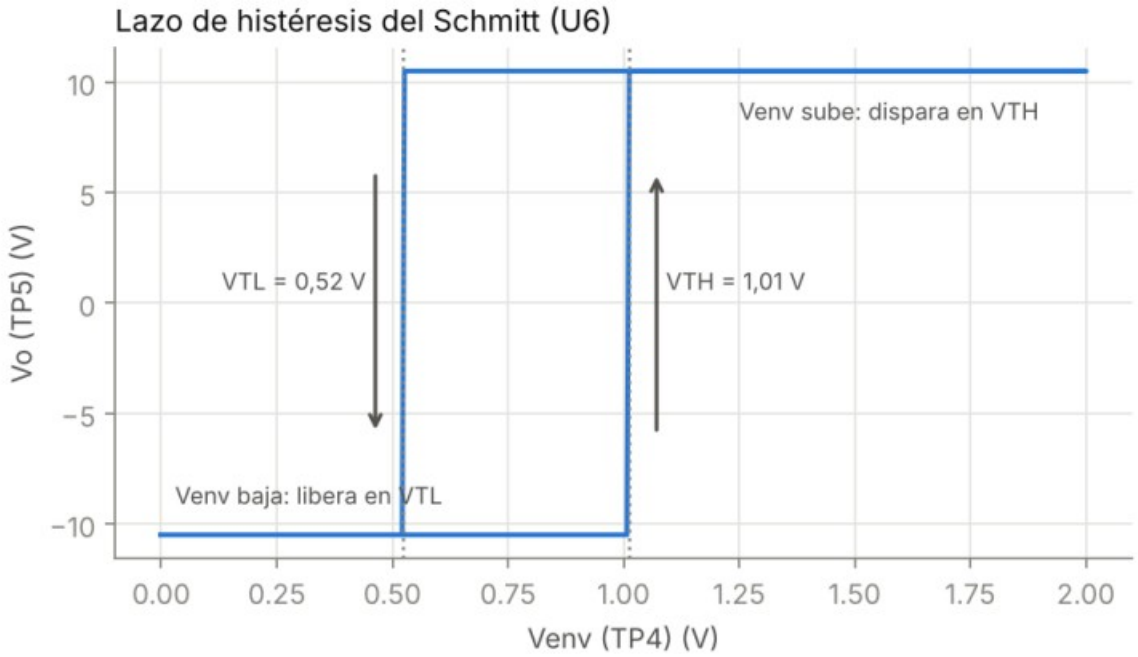
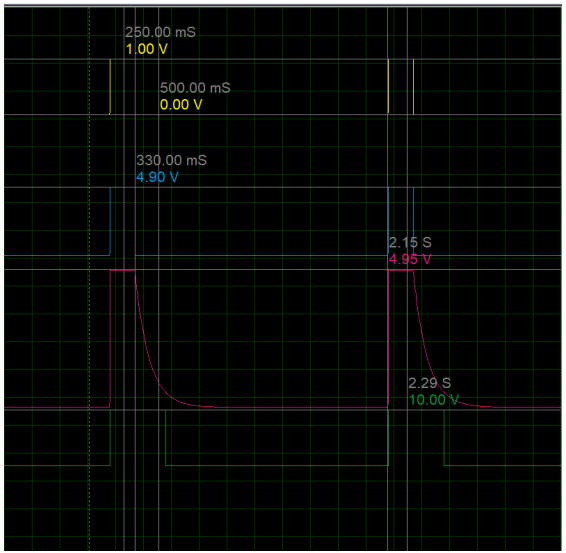
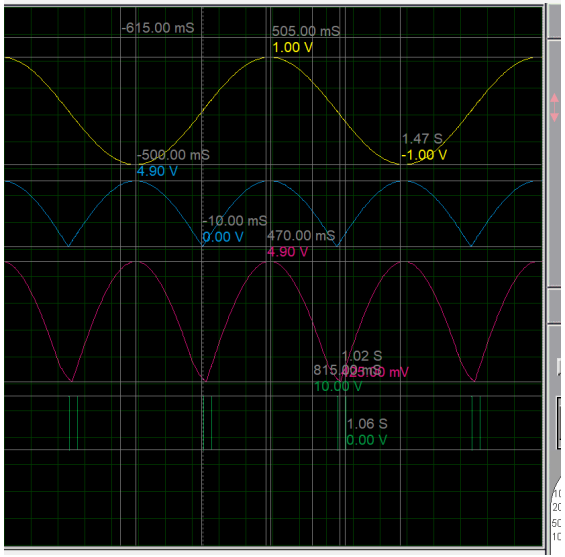



**Figura 10:** Lazo de histéresis del comparador: $V_o$ frente a $V_{env}$.

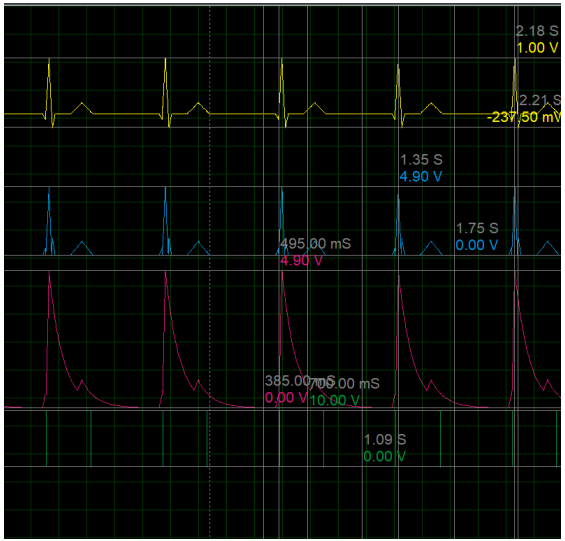
# Proteus simulation




**1 V pulse · 0.5 Hz**




**2 Vpp sine · 0.5 Hz**




**Synthetic heartbeat**

Yellow: input · Blue: Vr · Pink: Venv · Green: output

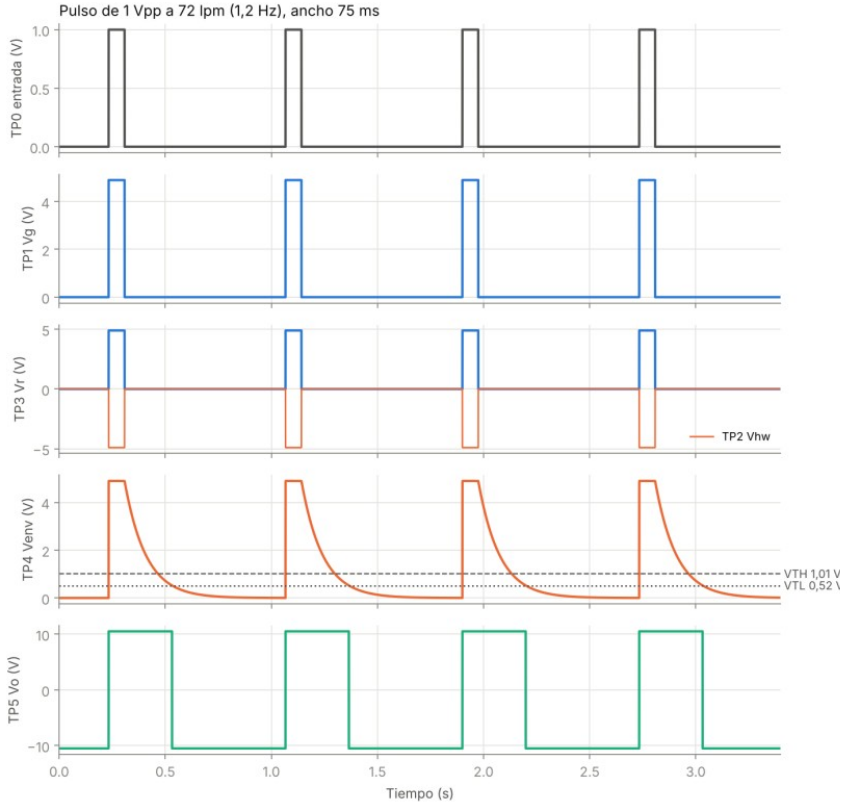



**Figura 13:** Formas de onda en TP0 a TP5 con un pulso de $1\,V_{pp}$ a 72 lpm.

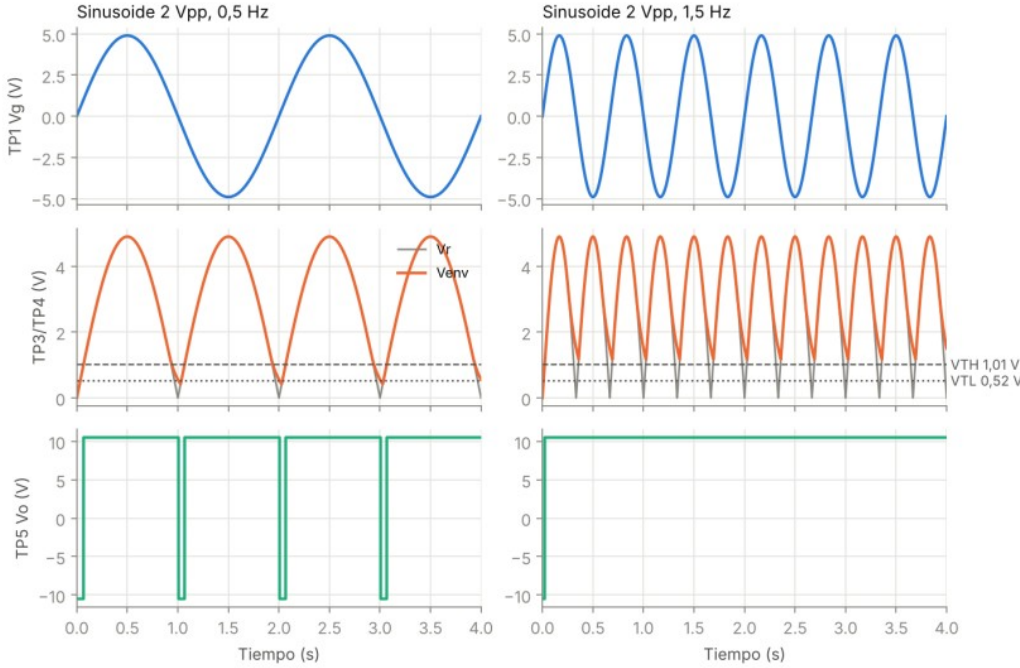



**Figura 14:** Sinusoide de $2\,V_{pp}$ a 0,5 Hz (dos pulsos por ciclo) y a 1,5 Hz (salida fija en alto).

# Simulation vs calculation

**4.90 V**

Vr peak (calc. 4.90 V)

**400 ms**

output width (calc. 405 ms)

**2**

pulses per sine cycle

**≤ 1.6 %**

maximum error

### Pending in the lab

- Measure τ, VTH, VTL and offset on the real build

# Conclusions

1. **Polarity** The precision rectifier detects both positive and negative R waves
2. **One pulse** The peak detector merges the QRS; $\tau = 100$ ms is the critical parameter
3. **No chatter** Schmitt hysteresis gives clean transitions
4. **Validated** Proteus matches the calculation (error $\leq 1.6$ %)

# References

https://github.com/Brahiam18/ecg-heartbeat-detector/tree/main